Syllabus

Statistical Mechanics. Phys-515, section 01.

Fall 2026.

Yurii Shylnov.

Sessions: Tuesday/Thursday. 11:15am-12:40pm. Classroom: TBA

Instructor: Prof. Yurii Shylnov.

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Office hours: MTW 12:45-1:50pm or by appointment.

Textbooks: R.K. Pathria, Paul D. Beale. Statistical Mechanics 4th Edition. Academic Press. 2021. 769 p. ISBN: 978-0081026922.

We are not going to follow the textbook chapter by chapter and not going to cover everything from these books.

Course Description:

1. Thermodynamics. Thermodynamic quantities, laws, and cycles. Ideal and non-ideal gas. Phases and phase transitions.
2. Classical statistical mechanics. Liouville theorem. Micro-canonical, canonical, and grand canonical ensembles. Partition function. Boltzmann and Maxwell distributions, and their applications.
3. Quantum statistics. Density matrix. Bose-Einstein and Fermi-Dirac distributions. Ideal quantum gases. Interacting Bose and Fermi systems. Quasiparticles formalism. Superfluidity, and superconductivity.
4. Quantum theory of phase transitions. Ising model, critical exponents, renormalization group. Basic ideas of non-equilibrium statistical mechanics.

Course Objectives

1. Apply thermodynamics laws to describe phenomena in the equilibrium systems.
2. Understand microcanonical, canonical, and grand canonical ensembles concept.
3. Use Boltzmann and Maxwell distributions to obtain properties of macroscopic systems.
4. Develop a quantum statistical mechanics.
5. Calculate macroscopic properties of various quantum systems.

Homework: You will receive your homework mostly every other week. All homework problems are to be done systematically and legibly. I am going to collect it and grade one problem selected randomly.

Project: It should be a mathematical problem of an acceptable level you find interesting and even exciting. Talk to me about your ideas ASAP. This is the key. After our conversation a title and abstract should be submitted by October 20th. A complete draft of a paper should be submitted by November 12th. A final version should be submitted by November 24th and discussed by December 3rd. Please, take this part of our course seriously and do not procrastinate.

Tests: There will be two tests, on October 8th, and December 1st. You are allowed to bring an equation sheet, A4 format, one side, handwritten by you personally to every test. **No final exams!**

There are no make-up tests, unless there are unexpected and well-documented emergency reasons and/or permission from the Dean of Students office. A make-up test will be slightly more complicated because a complete solution of the original test will be posted on Canvas.

Grading: Homework -30%, Tests - 40%, Project-30%.

Course Grade Determination: A 90-100%, B 80-89.9%, C 70-79.9%, D 60-69.9%, E below 60%.

Academic honesty: It is presumed that you will do your own work on the homework and tests. Discussing homework problems with others is encouraged but submitting work as your own which is copied or paraphrased from someone else is not permitted. Cheating includes, but not limited to, illegal collaboration, copying, using materials not permitted on tests, and aiding others on tests. Anyone found cheating will not be permitted to withdraw and will receive a grade of E for the course. Your academic dean will be informed and a statement will be placed in your permanent file. For more details, see the student handbook: https://www.iit.edu/student-affairs/student-handbook/fine-print/code-academic-honesty

Classroom environment Please, no cellular phones, iPods, iPads etc. in the class. I will be very much indebted to you if you do not eat, talk, text and so on in the classroom, because, believe it or not, it really distracts both me and your classmates. Of course, you are allowed to use your notebooks, tablets etc. to take notes but nothing else. Classes' format is going to be quite traditional. Lectures and discussions will take the most time.

Americans with Disabilities Act Policy Statement Reasonable accommodation will be made for students with documented disabilities. Contact the Center for Disability Resources to receive necessary support and help. The Center for Disability Resources (CDR) is located in 10 W. 35th St., Suite 3F3-1, phone: 312.567.5744 , e-mail: disabilities@illinoistech.edu

Title IX Compliance Title IX and our school policy prohibits discrimination on the basis of sex as well as all forms of sexual misconduct or harassment. I am always concerned about each and every student's well-being and ready to help. However, I am also required by the law to report any such incident to an Illinois Tech's Title IX coordinator. If you have experienced such an incident, please consult the following link for a list of resources available to you: https://web.iit.edu/hea-compliance/sexual-harassment-and-misconduct-prevention/title-ix-illinois-tech/resources.

USEFUL DATES FROM THE ACADEMIC CALENDAR

August 17	Courses Begin
September 7	Labor Day. No Classes.
October 8	Test # 1
October 12, 13	Fall Break Days. No Classes.
November 25-28	Thanksgiving Break. No Classes.
December 1	Test # 3
December 3	Last Day of Classes (Effective).
December 16	Final Grades are Due at Noon.